

Lab Program1 (b)

B) Consider the following database that maintain information about employees & Departments.

Employee (empid: int, ename: string, age:int, salary:int, #deptno:int)

Department (deptno:int, dname: string, #manager-id: int)

- Create the above tables by properly specifying the primary keys & foreign keys.
- Enter at least 5 tuples for each relation.
- Display emp-id & emp name whose salary lies between 10,000 and 50,000.
- List empname & salary for all the employee working for CS Dept.
- Display empname & deptname for all the managers.

i. Create the above tables by properly specifying the primary keys & foreign keys.

```
mysql> create database lab1_b;
Query OK, 1 row affected (0.00 sec)
```

```
mysql> use lab1_b;
Database changed
```

```
mysql> CREATE TABLE Employee (empid INT primary key, ename
-> VARCHAR(25) not null, age INT, salary INT default 0, deptno INT );
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> CREATE TABLE Department (deptno INT primary key, dname
-> VARCHAR(25) not null, mgr_id INT,
-> Foreign Key (mgr_id) references employee(empid));
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> ALTER TABLE Employee ADD CONSTRAINT fk_deptno FOREIGN
-> KEY (deptno) REFERENCES Department(deptno);
Query OK, 0 rows affected (0.08 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> desc employee;
```

Field	Type	Null	Key	Default	Extra
empid	int(11)	NO	PRI	NULL	
ename	varchar(25)	NO		NULL	
age	int(11)	YES		NULL	
salary	int(11)	YES		0	
deptno	int(11)	YES	MUL	NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> desc department;
```

Field	Type	Null	Key	Default	Extra
deptno	int(11)	NO	PRI	NULL	
dname	varchar(25)	NO		NULL	
mgr_id	int(11)	YES	MUL	NULL	

```
3 rows in set (0.01 sec)
```

ii. Enter at least 5 tuples for each relation.

```
mysql> insert into department(deptno,dname) values
```

```
-> (1,'cs'), (2,'accts'), (3,'sales'), (4,'admin'), (5,'purchase');
```

```
Query OK, 5 rows affected (0.03 sec)
```

```
Records: 5 Duplicates: 0 Warnings: 0
```

```
mysql> INSERT INTO Employee (empid, ename, age, salary, deptno)
```

```
-> VALUES (101, 'sachin', 35, 60000, 1);
```

```
Query OK, 1 row affected (0.03 sec)
```

```
mysql> INSERT INTO Employee VALUES
```

```
-> (102, 'sham', 40, 50000, 2);
```

```
Query OK, 1 row affected (0.01 sec)
```

```
mysql> INSERT INTO Employee VALUES
```

```
-> (103, 'ajay', 30, 55000, 2);
```

```
Query OK, 1 row affected (0.00 sec)
```

```
mysql> INSERT INTO Employee VALUES
```

```
-> (104, 'kapil', 27, 34000, 1);
```

```
Query OK, 1 row affected (0.03 sec)
```

```
mysql> INSERT INTO Employee VALUES
```

```
-> (105, 'rakesh', 32, 45000, 3);
```

```
Query OK, 1 row affected (0.05 sec)
```

```
mysql> update department set mgr_id =101 where deptno=1;
```

```
Query OK, 1 row affected (0.01 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> update department set mgr_id =101 where deptno=2;
```

```
Query OK, 1 row affected (0.02 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> update department set mgr_id =103 where deptno=3;
```

```
Query OK, 1 row affected (0.03 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> update department set mgr_id =104 where deptno=4;
```

```
Query OK, 1 row affected (0.03 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> update department set mgr_id =105 where deptno=5;
```

```
Query OK, 1 row affected (0.03 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT * from employee;
+-----+-----+-----+-----+-----+
| empid | ename  | age | salary | deptno |
+-----+-----+-----+-----+-----+
| 101 | sachin | 35 | 60000 | 1      |
| 102 | sham   | 40 | 50000 | 2      |
| 103 | ajay   | 30 | 55000 | 2      |
| 104 | kapil  | 27 | 34000 | 1      |
| 105 | rakesh | 32 | 45000 | 3      |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * from department;
+-----+-----+-----+
| deptno | dname  | mgr_id |
+-----+-----+-----+
| 1      | cs     | 101    |
| 2      | accts  | 101    |
| 3      | sales  | 103    |
| 4      | admin  | 104    |
| 5      | purchase | 105    |
+-----+-----+-----+
5 rows in set (0.00 sec)
```

iii. Display emp-id & emp name whose salary lies between 10,000 and 50,000.

```
mysql> SELECT empid, ename, salary FROM Employee WHERE salary
-> BETWEEN 10000 AND 50000;
+-----+-----+-----+
| empid | ename  | salary |
+-----+-----+-----+
| 102 | sham   | 50000 |
| 104 | kapil  | 34000 |
| 105 | rakesh | 45000 |
+-----+-----+-----+
3 rows in set (0.03 sec)
```

iv. List empname & salary for all the employee working for CS Dept.

```
mysql> SELECT e.ename,e. salary from employee e , department d
-> where d.dname='cs' and d.deptno=e.deptno;
+-----+-----+
| ename  | salary |
+-----+-----+
| sachin | 60000 |
| kapil  | 34000 |
+-----+-----+
2 rows in set (0.06 sec)
```

v. Display empname & deptname for all the managers.

```
mysql> SELECT e.ename, d.dname from employee e,department d  
-> where e.empid=d.mgr_id;
```

ename	dname
sachin	cs
sachin	accts
ajay	sales
kapil	admin
rakesh	purchase

5 rows in set (0.00 sec)